Customer Shopping Behavior Analysis

This presentation summarizes an analysis of 3,900 transactions to reveal spending patterns, customer segments, product preferences, subscription behavior, and actionable recommendations to guide business strategy.

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Project Overview

We analyzed transactional data from 3,900 purchases across multiple product categories. The goal: uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to inform strategic decisions across marketing, merchandising, and retention.

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Dataset Summary

Structure

Rows: 3,900 • Columns: 18. Key features include demographics, purchase details, and shopping behavior fields.

Missing Data

37 missing values in Review Rating. Missingness handled via imputation by category median during cleaning.

Key Fields

Customer demographics (age, gender, location, subscription), purchase details (item, category, amount, season, size, color), and behavioral signals (discounts, previous purchases, frequency, rating, shipping).

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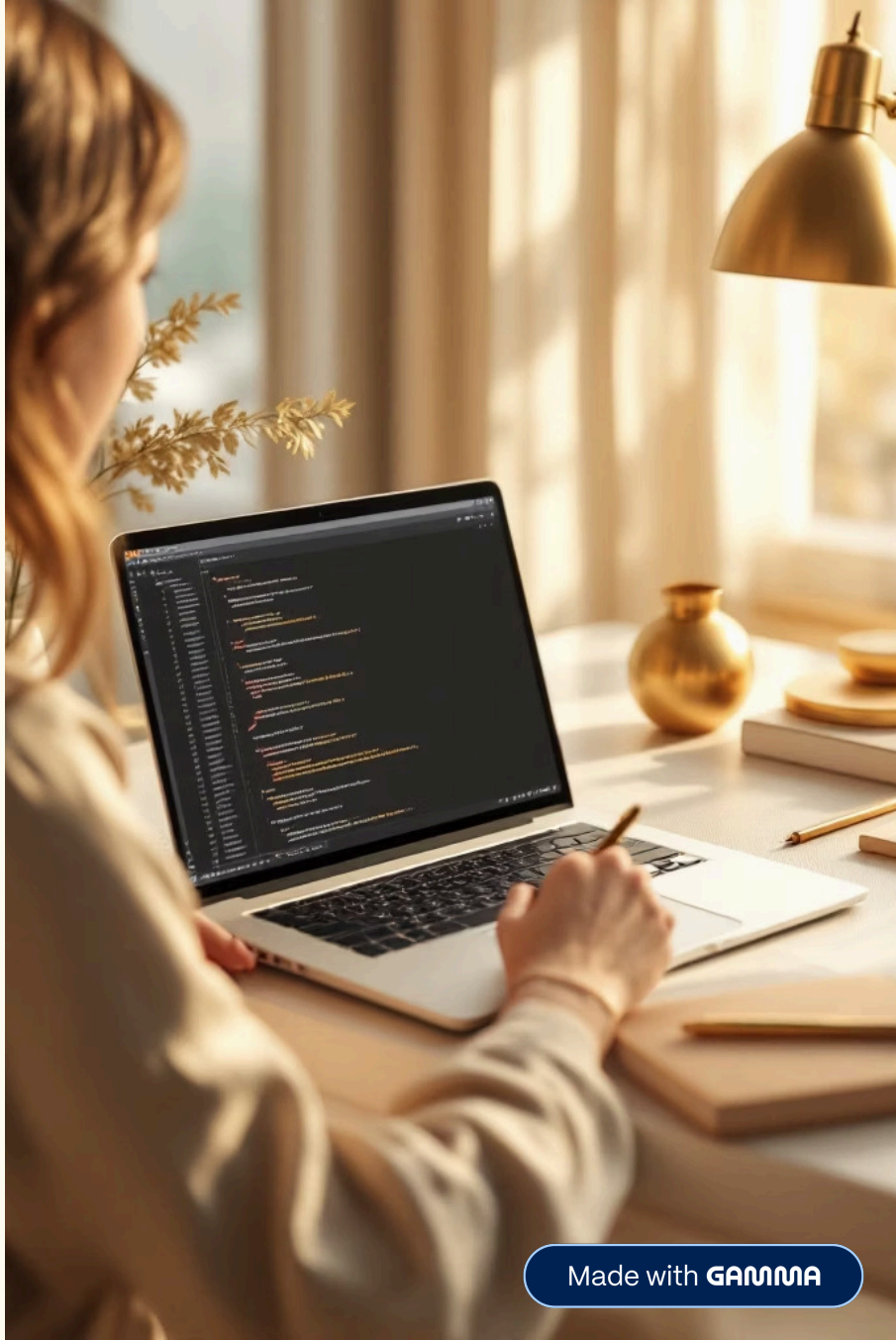
Exploratory Data Analysis — Python Workflow

Data loading with pandas, initial exploration via df.info() and .describe(), followed by cleaning and feature engineering. Actions included renaming columns to snake_case, binning age into age_group, deriving purchase_frequency_days, and checking redundancy between discount_applied and promo_code_used (promo_code_used was dropped).



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Data Cleaning & Preparation Details

Missing Ratings

Imputed Review Rating nulls using the median rating per product category to preserve distributional integrity.

Column Standardization

Renamed columns to snake_case for readability and consistent documentation across analysis scripts and SQL ingestion.

Feature Engineering

Added age_group bins and purchase_frequency_days, enabling segmentation and time-based behavior analysis.

Database Integration

Loaded cleaned DataFrame into PostgreSQL for structured SQL analytics and reporting.

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SQL Analysis — Key Business Queries

PostgreSQL queries answered core questions: revenue by gender, high-spending discount users, top products by rating, shipping comparisons, subscriber vs non-subscriber spend, discount-dependent products, customer segmentation, and top products per category.

Revenue by Gender

Female: 75,191 · Male: 157,890 — clear disparity to inform targeted campaigns.

Subscribers vs Non-Subscribers

Subscribers: 1,053 customers, avg spend $59.49, total revenue $62,645. Non-subscribers: 2,847 customers, avg spend $59.87, total revenue $170,436.

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Product & Discount Insights

Top Rated Items

Top 5 by average rating: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).

Discount-Dependent Products

Highest discount rates: Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37% — indicates pricing sensitivity by product.

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Behavioral Segments & Shipping

Customer segments: Loyal 3,116; Returning 701; New 83. Repeat buyer subscription check shows 958 repeat buyers subscribed vs. 2,518 non-subscribed repeat buyers. Shipping comparison: Standard avg purchase $58.46 vs Express $60.48 — slight premium for faster shipping.

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Dashboard Summary — Power BI

Dashboard highlights: 3.9K customers, $59.76 average purchase amount, 3.75 average review rating, subscription split Yes 27% / No 73%. Revenue by category: Clothing $100K, Accessories $70K, Footwear $30K, Outerwear $20K. Sales by category and age-group performance are included for quick decision-making.

Pricing Sensitivity
Top discounts: Hat 50%, Sneakers 49.66%

Customer Segments
Loyal 3,116; Returning 701; New 83

Subscription Insight
Repeat buyers: 958 subscribed, 2,518 not

Shipping Impact
Avg order: Standard $58.46, Express $60.48

Dashboard Summary
3.9K customers; avg $59.76; revenue by category

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits and targeted offers to increase subscriber conversion and lifetime value.

Customer Loyalty Programs

Reward repeat buyers to move customers into the Loyal segment and reduce churn.

Review Discount Policy

Balance promotional strategies with margin control — prioritize discounts on items with proven conversion lift.

Product Positioning

Highlight top-rated and best-selling products in campaigns to drive higher conversion and average order value.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users with personalized offers and timing to maximize ROI.

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